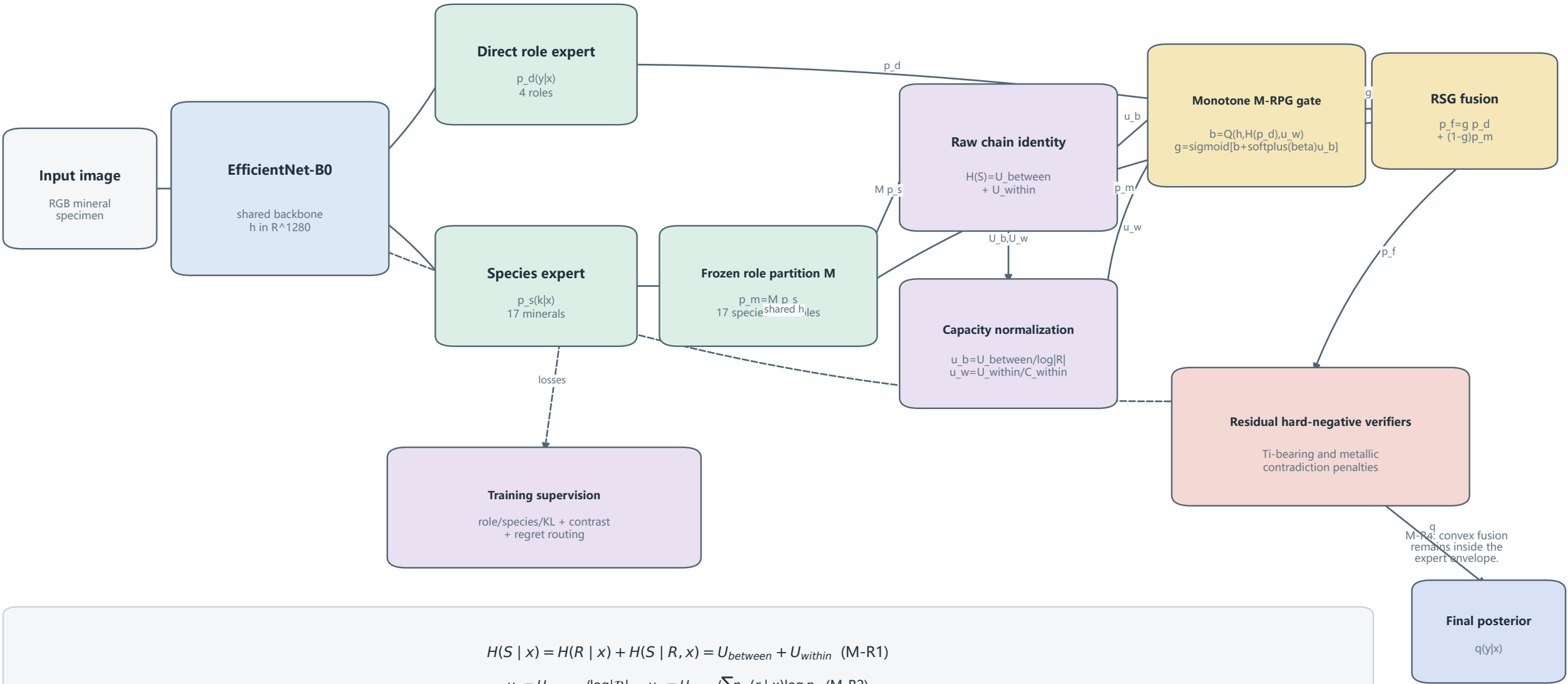


M-RPG-HRGV-Net: capacity-normalized uncertainty with monotone role-ambiguity routing

The frozen 17-to-4 mineral-role partition separates raw uncertainty, normalizes its available support, and constrains the role-ambiguity gate contribution.



$$H(S \mid x) = H(R \mid x) + H(S \mid R, x) = U_{\text{between}} + U_{\text{within}} \quad (\text{M-R1})$$

$$u_b = U_{\text{between}} / \log|R|, \quad u_w = U_{\text{within}} / \sum_r p_m(r \mid x) \log n_r \quad (\text{M-R2})$$

$$g = \sigma[b + \text{softplus}(\beta)u_b], \quad \partial g / \partial u_b = g(1 - g) \text{softplus}(\beta) \geq 0 \quad (\text{M-R3})$$